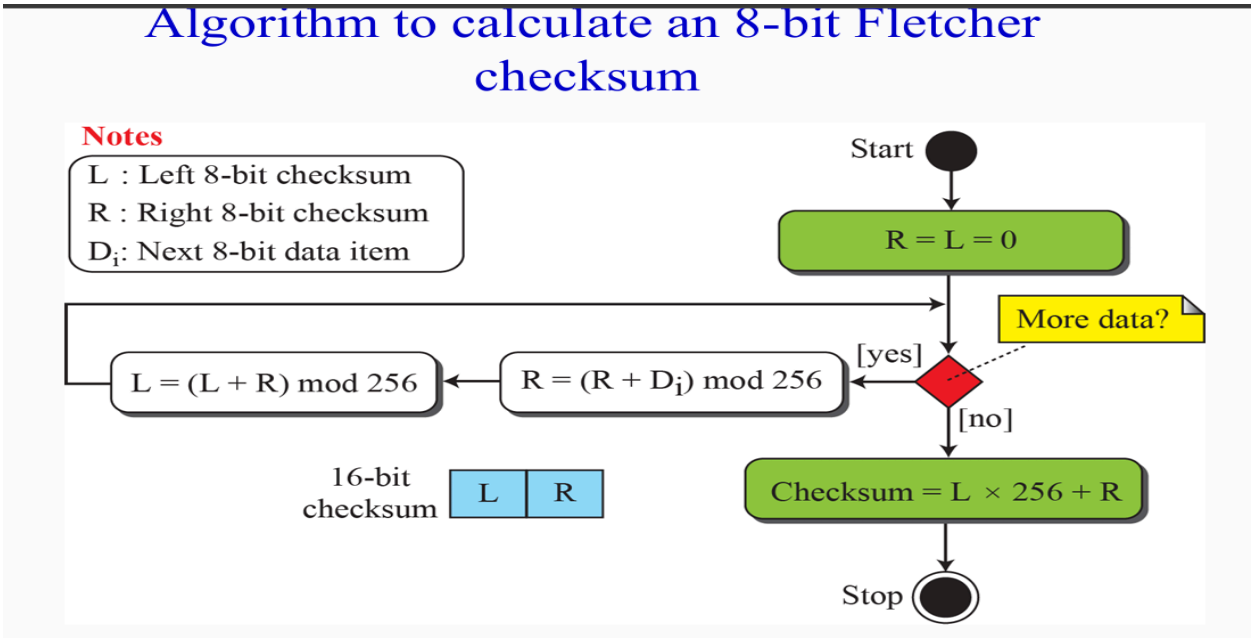
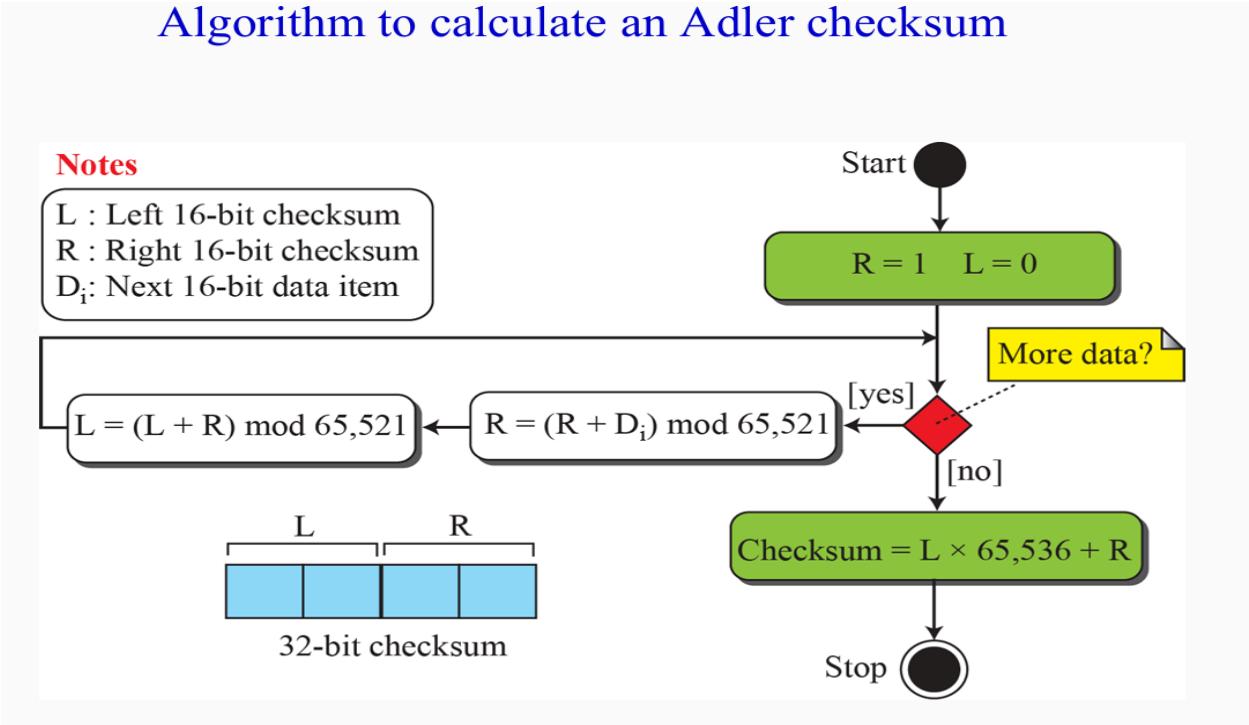
Q) Explain Fletcher checksum, and Adler checksum with flowchart. Calculate the checksum for the following bytes (2B)16,(3F)16,(6A)16 and (AF)16. using Fletcher algorithm, Also show that the result is a weighted checksum

Fletcher



Adler



Solve

Q. Manually simulate (Fletcher algo) to calculate the checksum $(2B)_{16}$ $(3F)_{16}$ $(6A)_{16}$ & $(AF)_{16}$ also show the result in a weighted checksum. First calculate R then L

Initial value $L=00$ $R=00$

item 1 $(2B)$

$L = L + R = 0 + 2B = 2B$

$R = R + D_i^0 = 0 + 2B = 2B$

item 2 $(3F)$

$L = 2B + 6A = 95$

$R = 2B + 3F = 6A$

item 3 $(6A)$

$L = 95 + D4 = 69$

$R = 6A + 6A = D4$

item 4 (AF)

$L = 69 + 83 = EC$

$R = D4 + AF = 83$

$\therefore$ checksum $= L \times 256 + R$

$= EC * 256 + 83$

$= EC * 100 + 83 = EC83$

$256 \rightarrow 16$ bit

Q) A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces a. 1000 frames per second? b. 500 frames per second? c. 250 frames per second?

